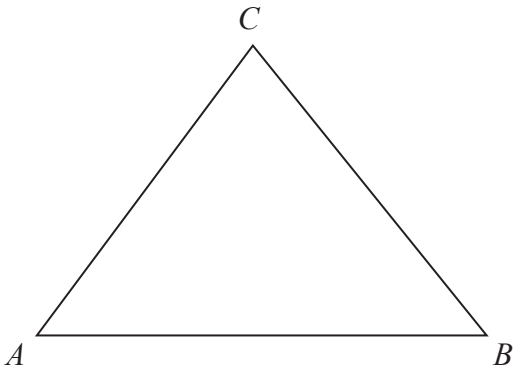
7 The diagram shows triangle *ABC*.



The perimeter of the triangle is 64 cm.

$AB : BC : CA = 6 : 5 : 5$

Work out the area of the triangle.

AB	BC	CA
6	5	5
×4	×4	×4
24	20	20

Half base = 24 ÷ 2 = 12 cm

$h^2 = 20^2 - 12^2 = 256$

$h = \sqrt{256} = 16$ cm

Area = ½ × 24 × 16

Area = 192 cm²

1 mark
Scale ratio by 64 ÷16 =4.

1 mark
Split isosceles triangle in half.

1 mark
Pythagoras gives height.

1 mark
½ × base × height.

1 mark
192 cm².

.....cm²

(Total for Question 7 is 5 marks)